Lionel Levine

Department of Mathematics, Cornell University

2023 – Professor
2019 – 2023 Frank Spitzer and Narahari Umanath Prabhu Associate Professor
2017 – 2019 Associate Professor
2011 – 2017 Assistant Professor

Department of Mathematics, MIT

2008 – 2011 C.L.E. Moore Instructor

Education

2002 – 2007 **Ph.D.** in Mathematics, University of California, Berkeley
1998 – 2002 **A.B.** in Mathematics, Harvard University

Funding

2026 – 2027 Coefficient Giving grant: Constitutional Evaluations and Character Training
2023 – 2025 Open Philanthropy grant: Long-term AI Safety Research
2015 – 2020 National Science Foundation CAREER Grant DMS-1455272
2014 – 2016 Alfred P. Sloan Research Fellowship
2011 – 2014 National Science Foundation Grants DMS-1105960 and DMS-1243606
2008 – 2011 National Science Foundation Postdoctoral Research Fellowship
2002 – 2007 National Science Foundation Graduate Research Fellowship

Awards

2020 Fellow of the American Mathematical Society
2018 – 2019 Simons Fellowship
2018 – 2019 Von Neumann Fellowship, Institute for Advanced Study
2014 Good Judgment Project "Superforecaster"
2013 Paul R. Halmos – Lester R. Ford Award, Mathematical Association of America

Research Interests

AI safety, probability, combinatorics, statistical physics

PhD students

Jonathn Chang
Suvadip Sana (PhD expected 2027; co-advised with Martin Wells)
Andrew Melchionna (PhD 2023)
Hannah Cairns (PhD 2022)
Feng Liang (PhD 2022)
Ryan McDermott (PhD 2021)
Lila Greco (PhD 2020)
Swee Hong Chan (PhD 2019)

Publications and Preprints

preprints 1. Lionel Levine, *Math for AI Safety: An invitation for mathematicians.*
London Mathematical Society Lecture Note Series, to appear. arXiv:2609.15289

2. Suvadip Sana, Sami Wolf, Neer Mehta, Alina Shah, Aitzaz Shaikh, Janna Goodman, and Lionel Levine, *Mixing times of Glauber dynamics on masked language models.* arXiv:2605.16378

3. Robin Kaiser, Lionel Levine, and Ecaterina Sava-Huss,
Locally Markov walks on finite graphs.
Random Structures & Algorithms, to appear. arXiv:2412.13766

2026 4. Jonathn Chang, Leonhard Piff, Suvadip Sana, Jasmine Li, and Lionel Levine,
EigenBench: A comparative behavioral measure of value alignment.
ICLR 2026 (oral presentation)

5. Bhagyesh Kumar, Ananya Sutradhar, Saurav Panigrahi, Jonathn Chang, and Lionel Levine,
Side effects of character training: Quantifying cross-constitution drift in LLMs.
Pluralistic Alignment Workshop at ICML 2026

2025 6. Viktor Kiss, Lionel Levine, and Lilla Tóthmérész,
The devil's staircase for chip-firing on random graphs and on graphons.
Random Structures & Algorithms (2025) 66, e21255.

7. Dan Hendrycks et al., *A definition of AGI.* arXiv:2510.18212

2024 8. Wilson Wu, John X. Morris, and Lionel Levine,
Do language models plan ahead for future tokens? CoLM 2024

9. Lionel Levine and Vittoria Silvestri,
Universality Conjectures for Activated Random Walk. Probability Surveys (2024) 21: 1-27

10. Lionel Levine and Feng Liang, *Exact sampling and fast mixing of Activated Random Walk.*
Electronic Journal of Probability (2024) 29:1–20

11. Elliot Glazer et al.,
FrontierMath: A benchmark for evaluating advanced mathematical reasoning in AI. arXiv:2411.04872

12. Rowan Hess and Lionel Levine,
How to quantify the coherence of a set of beliefs. arXiv:2412.02777

2023 13. Lila Greco and Lionel Levine, *Branching in a Markovian environment.*
Markov Processes and Related Fields (2023) v.29, Issue 1, 1–33

2022 14. Swee Hong Chan and Lionel Levine,
Abelian networks IV. Dynamics of nonhalting networks
Memoirs of the American Mathematical Society (2022) Volume 276 Number 1358

15. Lionel Levine, Hanbaek Lyu, and John Pike,
Double jump phase transition in a soliton cellular automaton.
International Math Research Notices (2022) 665–727

2021 16. Lionel Levine and Vittoria Silvestri,
How far do Activated Random Walkers spread from a single source?
Journal of Statistical Physics (2021) vol. 185 article 18

17. Swee Hong Chan, Lila Greco, Lionel Levine, and Peter Li, *Random walks with local memory.*
Journal of Statistical Physics (2021) vol. 184 article 6

2019 18. Shirshendu Ganguly, Lionel Levine, and Sourav Sarkar,
Formation of large-scale random structure by competitive erosion.
Annals of Probability (2019) 47:3649–3704

19. Daniel C. Jerison, Lionel Levine, and John Pike,
Mixing time and eigenvalues of the abelian sandpile Markov chain.
Transactions of the American Mathematical Society (2019) 372:8307–8345
20. Bob Hough, Daniel C. Jerison, and Lionel Levine, *Sandpiles on the square lattice.*
Communications in Mathematical Physics (2019) 367:33–87.
21. Lionel Levine and Vittoria Silvestri,
How long does it take for Internal DLA to forget its initial profile?
Probability Theory and Related Fields (2019) 174:1219–1271.
22. Alexander E. Holroyd, Lionel Levine, and Peter Winkler, *Abelian logic gates.*
Combinatorics, Probability, and Computing (2019) 28:388–422.

2018 23. Wilfried Huss, Lionel Levine and Ecaterina Sava-Huss,
Interpolating between random walk and rotor walk.
Random Structures & Algorithms (2018) 52.2:263–282

2017 24. Lionel Levine, Wesley Pegden and Charles K. Smart,
The Apollonian structure of integer superharmonic matrices.
Annals of Math (2017) 186:1–67
25. Elisabetta Candellero, Shirshendu Ganguly, Christopher Hoffman and Lionel Levine,
Oil and water: a two-type internal aggregation model.
Annals of Probability (2017) 45:4019–4070
26. Lionel Levine and Yuval Peres, *Laplacian growth, sandpiles and scaling limits.*
Bulletin of the American Mathematical Society (2017) 54:355–382
27. Lionel Levine and Ramis Movassagh,
The gap of the area-weighted Motzkin spin chain is exponentially small.
Journal of Physics A: Mathematical and Theoretical (2017) 50:255302
28. Shirshendu Ganguly, Lionel Levine, Yuval Peres and James G. Propp,
Formation of an interface by competitive erosion.
Probability Theory and Related Fields (2017) 168:455–509

2016 29. Benjamin Bond and Lionel Levine, *Abelian networks I. Foundations and examples.*
SIAM Journal on Discrete Mathematics (2016) 30:856–874.
30. Benjamin Bond and Lionel Levine, *Abelian networks II. Halting on all inputs.*
Selecta Mathematica (2016) 22:319–340.
31. Benjamin Bond and Lionel Levine, *Abelian networks III. The critical group.*
Journal of Algebraic Combinatorics (2016) 43:635–663.
32. Laura Florescu, Lionel Levine and Yuval Peres, *The range of a rotor walk.*
The American Mathematical Monthly (2016) 123(7):627–642.
33. Matthew Farrell and Lionel Levine, *CoEulerian graphs.*
Proceedings of the American Mathematical Society (2016) 144:2847–2860.
34. Matthew Farrell and Lionel Levine, *Multi-Eulerian tours of directed graphs.*
Electronic Journal of Combinatorics (2016) 23:P2.21.
35. Lionel Levine, Mathav Murugan, Yuval Peres and Baris Ugurcan, *The divisible sandpile at critical density.* Annales Henri Poincaré (2016) 17(7):1677–1711.
36. Lionel Levine, Wesley Pegden and Charles K. Smart, *Apollonian structure in the abelian sandpile.* Geometric And Functional Analysis (2016) 26(1):306–336.

2015 37. Lionel Levine, *Threshold state and a conjecture of Poghosyan, Poghosyan, Priezzhev and Ruelle.*
Communications in Mathematical Physics (2015) 335(2):1003–1017

38. Louis J. Billera, Lionel Levine and Karola Mészáros, *How to decompose a permutation into a pair of labeled Dyck paths by playing a game.*
Proceedings of the American Mathematical Society (2015) 143:1865–1873.

2014 39. Lionel Levine and Yuval Peres, *The looping constant of $\mathbb{Z}^d$.*
Random Structures & Algorithms (2014) 45:1–13

40. David Jerison, Lionel Levine and Scott Sheffield, *Internal DLA and the Gaussian free field.*
Duke Mathematical Journal (2014) 163(2):267–308

41. Laura Florescu, Shirshendu Ganguly, Lionel Levine and Yuval Peres, *Escape rates for rotor walks in $\mathbb{Z}^d$.* SIAM Journal on Discrete Mathematics (2014) 28(1):323–334.

42. David Jerison, Lionel Levine, and Scott Sheffield. *Internal DLA for cylinders*, in *Advances in Analysis: The Legacy of Elias M. Stein*,
Princeton University Press (2014) 189.

2013 43. Lionel Levine, Scott Sheffield and Katherine E. Stange, *A duality principle for selection games.*
Proceedings of the American Mathematical Society (2013) 141(12):4349–4356.

44. David Jerison, Lionel Levine and Scott Sheffield, *Internal DLA in higher dimensions.*
Electronic Journal of Probability (2013) 18(98):1–14.

45. Tobias Friedrich and Lionel Levine, *Fast simulation of large-scale growth models.*
Random Structures & Algorithms (2013) 42:185–213.

46. Christopher J. Hillar, Lionel Levine and Darren Rhea, *Equations solvable by radicals in a uniquely divisible group.* Bulletin of the London Mathematical Society (2013) 45:61–79.

2012 47. David Jerison, Lionel Levine and Scott Sheffield, *Logarithmic fluctuations for internal DLA.*
Journal of the American Mathematical Society (2012) 25:271–301.

48. Lionel Levine and Katherine E. Stange, *How to make the most of a shared meal: plan the last bite first.* American Mathematical Monthly (2012) 119:550–565.

49. Giuliano Giacaglia, Lionel Levine, James Propp and Linda Zayas-Palmer.
Local-to-global principles for the hitting sequence of a rotor walk.
Electronic Journal of Combinatorics (2012) 19:P5.

2011 50. Lionel Levine, *Sandpile groups and spanning trees of directed line graphs.*
Journal of Combinatorial Theory A (2011) 118:350–364.

51. Lionel Levine, *Parallel chip-firing on the complete graph: devil's staircase and Poincaré rotation number.* Ergodic Theory and Dynamical Systems (2011) 31:891–910

2010 52. Anne Fey, Lionel Levine and David B. Wilson, *Driving sandpiles to criticality and beyond.*
Physical Review Letters (2010) 104:145703.

53. Anne Fey, Lionel Levine and David B. Wilson, *The approach to criticality in sandpiles.*
Physical Review E (2010) 82:031121.

54. Anne Fey, Lionel Levine and Yuval Peres, *Growth rates and explosions in sandpiles.*
Journal of Statistical Physics (2010) 138:143–159.

55. Lionel Levine and Yuval Peres, *Scaling limits for internal aggregation models with multiple sources.* Journal d'Analyse Mathématique (2010) 111:151–219.

56. Lionel Levine and James Propp, *What is a sandpile?*
Notices of the American Mathematical Society (2010) 57:976–979.

57. Wouter Kager and Lionel Levine, *Rotor-router aggregation on the layered square lattice.*
Electronic Journal of Combinatorics (2010) 17:R152.

58. Wouter Kager and Lionel Levine, *Diamond aggregation.*
Mathematical Proceedings of the Cambridge Philosophical Society (2010) 149:351–372.

2009 59. Lionel Levine and Yuval Peres, *Strong spherical asymptotics for rotor-router aggregation and the divisible sandpile.* Potential Analysis (2009) 30:1–27.

60. Itamar Landau and Lionel Levine, *The rotor-router model on regular trees.*
Journal of Combinatorial Theory A (2009) 116:421–433.

61. Lionel Levine, *The sandpile group of a tree.*
European Journal of Combinatorics (2009) 30:1026–1035.

2008 62. Alexander E. Holroyd, Lionel Levine, Karola Mészáros, Yuval Peres, James Propp and David B. Wilson, *Chip-firing and rotor-routing on directed graphs*, in "In and Out of Equilibrium 2," Progress in Probability vol. 60, 331–364.

63. Lionel Levine and Yuval Peres, *Spherical asymptotics for the rotor-router model in* $\mathbb{Z}^d$.
Indiana University Mathematics Journal (2008) 57:431–450.

2007 64. Christopher J. Hillar and Lionel Levine, *Polynomial recurrences and cyclic resultants.*
Proceedings of the American Mathematical Society (2007) 135:1607–1618.

2006 65. Lionel Levine, *Fractal sequences and restricted Nim.*
Ars Combinatoria (2006) 80:113–127.

Selected Talks

2026 Nov. University of Pennsylvania mathematics colloquium
Oct. Maths × AI Safety workshop, Safe AI Germany, Bonn (remote talk)
Oct. Fields Institute Mathematical AI Seminar, Toronto
Oct. Aspen Meeting on Foundation Models
Sep. Northwestern mathematics colloquium
Aug. OpenAI Math Summit
Jul. Budapest Semesters in Mathematics colloquium, Budapest, Hungary
Apr. UT Austin AI alignment seminar
Feb. Cornell Global AI seminar
2025 Nov. Forethought research seminar
Oct. Cornell Statistics Graduate Society
Sep. Ohio State mathematics colloquium
2024 Nov. Cornell Physics Colloquium
Oct. Isaac Newton Institute, Cambridge, UK (minicourse)
Aug. Cornell Oliver Club
Feb. University of Miami mathematics colloquium
2023 Dec. Online Seminar on Undergraduate Mathematics Education (OLSUME)
Apr. La Sapienza probability seminar, Rome, Italy
Feb. Alignment Workshop, OpenAI, San Francisco
2022 Dec. SPCS2022 in honor of Deepak Dhar receiving the Boltzmann Medal
2021 Oct. Berkeley probability seminar
Apr. University of Washington probability seminar
Mar. Duke probability seminar
Mar. University of Birmingham applied math seminar

2020 Oct. One World probability seminar
Oct. AMS Sectional meeting (two talks)
2019 Jun. Joint Meeting of the Vietnam and American Mathematical Societies, Quy Nhon, Vietnam
Mar. IAS discrete math seminar
Mar. University of Chicago probability seminar
Feb. ICTS program on universality in random structures, Bangalore, India (4 lectures)
Feb. IIT Mumbai mathematics colloquium
2018 Sep. Penn State MASS seminar and mathematics colloquium
Jul. Montreal workshop on challenges in probability and statistical physics
2017 Oct. Stony Brook mathematics colloquium
Oct. Princeton probability seminar
Sep. Cornell probability and dynamics (joint seminar)
Jul. Mathematical Congress of the Americas, Special session on probability, Montreal, Canada
2016 Nov. Berkeley probability seminar
Apr. Finger Lakes Probability Seminar
Jan. CIRM workshop on nonequilibrium statistical physics, Marseille, France
2015 Oct. CIB Conference on statistical physics on transitive graphs, EPFL Lausanne, Switzerland
Aug. MAA Centennial, Invited session on "Arithmetic of Spheres"
Apr. Harvard random matrix and statistical physics seminar
Feb. Triangle Lectures in Combinatorics
2014 Dec. Tufts mathematics colloquium
Jun. CIRM Conference on Random Media, Marseille, France
May MIT probability seminar
Apr. University of Arizona mathematics colloquium
2013 Nov. UPenn combinatorics and probability seminar
Oct. Midwest Probability Colloquium
Aug. Mathematical Congress of the Americas, special session on applied combinatorics
Jan. Cornell applied mathematics colloquium
2012 Dec. Tel Aviv University mathematics colloquium
Dec. Weizmann Institute geometric functional analysis and probability seminar
Apr. University of Connecticut mathematics colloquium
2011 Nov. 80th birthday conference in honor of Harry Kesten
Mar. MSRI workshop on free boundary problems
Jan. University of British Columbia mathematics colloquium
2010 Feb. IMPA probability seminar, Rio de Janeiro, Brazil
Jan. University of Washington / PIMS mathematics colloquium
2009 Oct. Dartmouth computer science colloquium
Jul. 27th Brazilian Colloquium of Mathematics, Rio de Janeiro
2008 Aug. CRM Workshop on Laplacian growth and related topics, Montreal
2007 Oct. International symposium on stochastic large scale interacting systems, Fukuoka, Japan

Professional Service

- Forecaster, Longitudinal Expert AI Panel (LEAP), Forecasting Research Institute (2026–).
- Editor, *Combinatorial Theory* (2020–2025).
- Co-organizer, Cornell Probability Summer School (2013, 2014, 2019, 2022).
- Co-organizer, Invited session at the World Congress in Probability and Statistics, July 2016.
- Co-organizer, AIM workshop on generalizations of chip-firing and the critical group, July 2013.
- Co-organizer, Cornell probability seminar (2012–2026)
- Co-organizer, MIT probability seminar (2008–2011)

Journals and Conferences Refereed

Advances in Mathematics
American Mathematical Monthly
Annals of Combinatorics
Annals of Probability
Archive for Rational Mechanics and Analysis
Combinatorics, Probability and Computing
Communications in Mathematical Physics
Discrete Mathematics
Duke Mathematical Journal
Electronic Journal of Combinatorics
Foundations of Computer Science (**FOCS**)
Geometric and Functional Analysis
International Mathematics Research Notices
Journal d'Analyse Mathématique
Journal of the American Mathematical Society
Journal of Combinatorial Theory A
Journal of Integer Sequences
Journal of Physics A
Journal of Statistical Physics
Linear Algebra and Its Applications
Mathematics of Operations Research
Notices of the American Mathematical Society
Physica A
Physical Review E
Potential Analysis
Probability Theory and Related Fields
Random Structures & Algorithms
SIAM Journal on Discrete Mathematics
Stochastic Processes and Their Applications
Symposium on Discrete Algorithms (**SODA**)
Symposium on Theory of Computing (**STOC**)

Teaching

2026	Undergraduate Probability (MATH 4710)
2025	Undergraduate Probability (MATH 4710)
	Graduate Probability I (MATH 6710)

	Graduate Probability II (MATH 6720)
2024	Math for AI Safety (MATH 7710, graduate topics class in probability)
2023	Strategy, Cooperation, and Conflict (MATH 1340)
2022	Graduate Probability I (MATH 6710)
	Limits of discrete random structures (MATH 7710, graduate topics class in probability)
2021	Graduate Probability I (MATH 6710)
	Calculus I (MATH 1110)
	Graduate Probability II (MATH 6720)
2020	Abelian Networks (MATH 7710, graduate topics class in probability)
	Graduate Probability I (MATH 6710)
	Nonlinear Dynamics and Chaos (MATH 4210/ MAE 5790)
	Graduate Probability II (MATH 6720)
2019	Graduate Probability I (MATH 6710)
2018	Graduate Probability II (MATH 6720)
2017	Graduate Probability II (MATH 6720)
	Stochastic Processes (MATH 4740)
2016	Mathematics and Politics (MATH 1340)
2015	Stochastic Processes (MATH 4740)
2014	Graduate Probability I (MATH 6710)
	Stochastic Processes (MATH 4740)
2013	Stochastic Processes (MATH 4740)
2012	Laplacian Growth (MATH 7770, graduate topics class in probability)
2011	Algebraic Combinatorics (18.312) at MIT

Mentoring and Outreach

2026	Mentor for Jinzhou Wu in the Pivotal Fellowship extension
	Judge for the Google DeepMind Hackathon "Measuring Progress toward AGI"
	Co-organizer of the Cornell Math+AI Lab
2025–2026	Faculty advisor of CAIA (Cornell AI Alignment student club)
2025	Mentored undergraduates Janna Goodman and Neer Mehta on *Glauber dynamics on a natural language state space*.
2024 & 2026	Mentor for SPAR (Supervised Program for Alignment Research)
2023	Mentored undergraduate Oliver Hopcroft on *eliciting from language models*.
2022	Mentored undergraduates Rowan Hess and Luchen Shi on *rectifying inconsistent priors*; Aidan O'Gara on *eliciting from language models*; Lukas Berglund on *deceptive alignment*; Katrina Brown and Aditya Syam on *multi-agent reinforcement learning*.
2021	Mentored undergraduate Bryan Lu on a *stochastic model of attention*.
2019	Taught a five-week minicourse on *probability, combinatorics, and computation* for gifted high school students at the PROMYS program.
	Designed a research project on *ladder sandpiles* for PROMYS students Siddhant Chaudhary and Ashwin Padaki. They presented at the 2020 Joint Math Meetings.

2018	Designed research projects on *random spanning trees* and *greedy avoidance of arithmetic progressions* for PROMYS undergraduates and high school students
	Spoke on *the future of prediction* at the MAA New York sectional meeting
2017	Supervised senior thesis research for Cornell math major Peter Li. Peter's thesis topic: *The vector-valued martingale invariance principle*
	Designed a research project on *random game trees* for PROMYS students
2016	PROMYS alumni lecture, Boston University, July 6, 2016.
	Math Awareness Public Lecture, Cornell, April 29, 2016.
2015	Designed a research project on *box-ball solitons* for PROMYS students Karthik Karnik and Mikaeel Yunus, and sponsored them for the Siemens and JSHS competitions.
2014	Supervised senior thesis research for Cornell undergraduate Matthew Farrell. Matt's thesis title: *The halting problem for chip-firing on finite directed graphs*
	Supervised research for Cornell undergraduates Young Jun Song and Matt Weatherly.
2013	Supervised senior thesis research for Cornell undergraduate Diwakar Raisingh. Diwakar's thesis title: *Toward an axiomatic characterization of the smash sum*
	Presented on *the sandpile group of a graph* at the Cornell Math Club.
2012	Presented on *patterns in coin-flipping* to an audience of secondary school teachers.
	Contributed mathematical art to MSRI's Wild Beauty outreach event.
2011	Presented on *primes in Pascal's triangle* at the Boston Math Circle.
	Co-mentored high school student Ziv Scully in MIT's PRIMES program.
2009–2011	Mentored MIT undergraduates Joshua Alman, Ben Bond, Giuliano Giacaglia, Neil Gurram, Damien Jiang, Aviva Siegel and Linda Zayas-Palmer in the UROP program.